

Direct Offline Music Deployment to Samsung Galaxy Watch (SM-R800) from ChromeOS Flex Using SDB over Wi-Fi

A Reproducible Technical Whitepaper for Linux and ChromeOS Users

Tejakiran Naik S

June 2026

Abstract

Samsung's Galaxy Watch SM-R800 was designed around a smartphone-centric ecosystem. Official workflows generally require a companion mobile device for media synchronization.

This paper presents a reproducible methodology for transferring, indexing, validating, and playing offline music directly on a Samsung Galaxy Watch SM-R800 using ChromeOS Flex, Linux, and Samsung Secure Debug Bridge (SDB) over Wi-Fi.

The work documents network architecture, filesystem investigation, media database validation, troubleshooting procedures, and automation opportunities. The resulting workflow eliminates smartphone dependency and extends the useful lifespan of legacy Tizen-based smartwatches.

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1 Introduction

Samsung Galaxy Watch devices running Tizen OS remain widely used despite Samsung’s transition toward Wear OS.

Many users continue to operate these devices without access to Samsung’s official smartphone ecosystem. Existing documentation primarily assumes the presence of:

- Samsung Galaxy Wearable
- Android smartphone
- Official synchronization workflows

This paper investigates whether direct music deployment can be achieved from a Linux environment without a smartphone.

2 Research Objectives

The primary objectives are:

1. Establish wireless communication between ChromeOS Flex and Galaxy Watch.
2. Transfer audio files directly using SDB.
3. Validate filesystem placement.
4. Verify media database indexing.
5. Confirm successful playback.
6. Document a reproducible workflow.

3 Experimental Environment

3.1 Host System

Component	Specification
Operating System	ChromeOS Flex
Linux Environment	Penguin Container
Architecture	x86_64
Toolchain	Tizen Studio
Communication	Wi-Fi

Table 1: Host Environment

3.2 Target Device

Component	Specification
Device	Samsung Galaxy Watch
Model	SM-R800
Operating System	Tizen OS
Storage	Internal Flash Storage
Connectivity	Wi-Fi

Table 2: Target Device

4 System Architecture

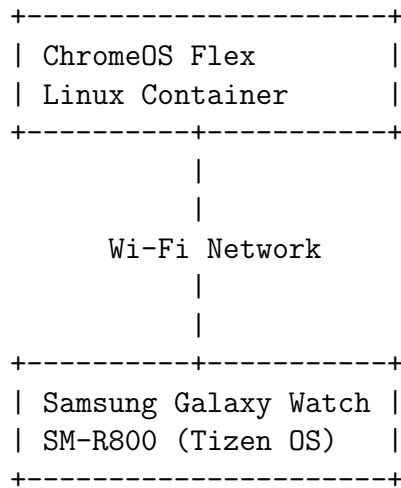


Figure 1: High-Level System Architecture

5 Network Communication Model

The watch and host machine must reside on the same subnet.

Connectivity is verified using ICMP echo requests before SDB communication begins.

5.1 Connectivity Test

```
ping 192.168.31.86
```

Expected result:

```
64 bytes from 192.168.31.86
```

6 Developer Configuration

The following settings must be enabled:

1. Developer Mode
2. Debugging
3. Wi-Fi Debugging

The watch IP address is obtained from Wi-Fi settings.

7 SDB Communication

7.1 Connection

```
~/tizen-studio/tools/sdb connect 192.168.31.86:26101
```

7.2 Verification

```
~/tizen-studio/tools/sdb devices
```

Expected output:

```
List of devices attached
192.168.31.86:26101 device
```

8 Filesystem Investigation

8.1 Music Directory

```
/opt/usr/media/Music/
```

8.2 Media Database

```
/opt/usr/home/owner/.applications/dbspace/.media.db
```

9 Music Deployment Workflow

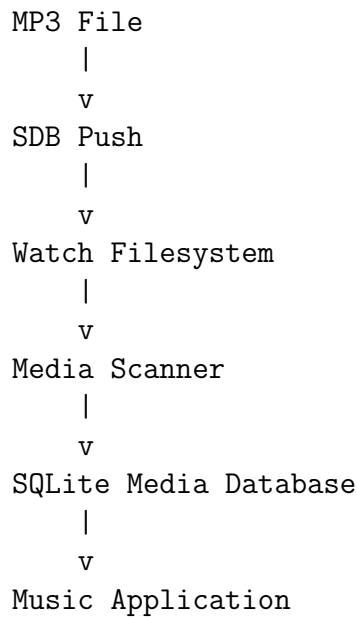


Figure 2: Music Deployment Pipeline

9.1 File Transfer

```
~/tizen-studio/tools/sdb push song.mp3 \  
/opt/usr/media/Music/
```

10 Media Database Validation

10.1 Audio Count Query

```
~/tizen-studio/tools/sdb shell sqlite3 \  
/opt/usr/home/owner/.applications/dbspace/.media.db \  
"SELECT count(*) FROM media WHERE media_type=3;"
```

Media type 3 corresponds to audio files.

11 Experimental Results

Test	Result
Ping Connectivity	Pass
SDB Connection	Pass
File Transfer	Pass
Database Detection	Pass
Music Recognition	Pass
Playback	Pass

Table 3: Experimental Results

12 Troubleshooting Matrix

Problem	Cause	Resolution
No Devices Found	Debugging Disabled	Re-enable Debugging
Connection Refused	Incorrect IP	Verify IP Address
Transfer Failure	Network Interruption	Reconnect Wi-Fi
Missing Music	Scanner Delay	Reboot Watch
Metadata Missing	Corrupt Tags	Repair Metadata

13 Security Considerations

Developer mode increases the attack surface of the device.

Recommended practices:

- Disable debugging after transfers.
- Use trusted networks.
- Avoid public Wi-Fi.
- Restrict access to development systems.

14 Future Work

Potential extensions include:

- Playlist synchronization
- Album artwork support
- Automatic deployment scripts
- Metadata repair tools
- Backup and restore workflows

15 Conclusion

This study demonstrates that Samsung Galaxy Watch SM-R800 devices can function as standalone offline music players without smartphone synchronization.

Using SDB over Wi-Fi, ChromeOS Flex users can directly transfer, validate, and play music through a reproducible workflow.

The methodology extends the useful lifespan of legacy Tizen devices and provides a practical alternative to proprietary synchronization systems.

Appendix A: Command Reference

```
ping <WATCH_IP>
```

```
~/tizen-studio/tools/sdb connect <WATCH_IP>:26101
```

```
~/tizen-studio/tools/sdb devices
```

```
~/tizen-studio/tools/sdb push song.mp3 \  
/opt/usr/media/Music/
```

```
~/tizen-studio/tools/sdb shell sqlite3 \  
/opt/usr/home/owner/.applications/dbspace/.media.db \  
"SELECT count(*) FROM media WHERE media_type=3;"
```